

Cataract patterns and functional effect

Lens opacity causes gradual painless blur, but pattern and daily impact guide interpretation.

Locate the opacity

Pattern	Clinical link
Nuclear sclerosis	Yellow-brown nucleus; index myopia and possible second sight.
Cortical cataract	Radial spokes; glare can be prominent.
Posterior subcapsular cataract	Posterior central opacity; steroid association and near difficulty.
Posterior capsule opacification	Late recurrent blur after cataract extraction; treated with Nd:YAG capsulotomy.

Treat the visual limitation

Confirm that cataract is the important cause of the complaint before discussing surgery. Difficulty with daily activities and glare matter alongside acuity; a single chart threshold is not an absolute rule. Refraction may help temporarily, but there is no definitive medicine that clears a cataract. Cataract extraction replaces the opaque lens with an intraocular lens.

Read the lid anatomy

A lump, malposition or abnormal lid margin can each damage comfort and ocular function.

Appearance suggests the process

Finding	Likely process
Acute tender lash-base pustule	External hordeolum: infected Zeiss or Moll gland.
Slow firm painless tarsal lump	Chalazion: meibomian lipogranuloma.
Capped meibomian openings; thick secretions	Posterior blepharitis / meibomian gland dysfunction.
Ulceration, lash loss or recurrent atypical lump	Assess for lid malignancy.

Lid position also matters

Entropion rotates the margin inward; trichiasis has misdirected lashes with the lid normally positioned. Ectropion turns the lid and punctum outward, causing exposure and impaired drainage. Persistent irritation requires assessment of the corneal surface. In a child, ptosis that blocks the visual axis is a threat to visual development rather than only an appearance concern.

References Ashley card 1699860546135; Ashley card 1699861730272; Ashley card 1699849822997; Ashley card 1699843989909

Watering: distinguish gland from drainage

The site of swelling helps separate dacryoadenitis from lacrimal sac disease.

Site and accompanying clues

Pattern	Interpretation
Superotemporal swelling; S-shaped upper lid	Dacryoadenitis affects the lacrimal gland.
Painful swelling below medial canthus; pus reflux	Dacryocystitis affects the lacrimal sac.
Infant watering; clear cornea; delayed dye clearance	Congenital nasolacrimal obstruction is likely.
Everted lower punctum	Ectropion can prevent tears entering the drainage system.

Do not treat every watery eye alike

Uncomplicated infant nasolacrimal obstruction may improve with massage and review. Painful sac swelling or fever suggests infection rather than uncomplicated obstruction. Photophobia, corneal haze or enlargement in an infant requires assessment for other ocular disease. Fluorescein dye disappearance helps assess drainage but does not itself establish infection or its exact anatomical cause.

Protect vision during development

A clear retinal image and appropriate use of each eye are essential in childhood.

Three examination priorities

Acuity and refraction

Unequal best-corrected acuity needs organic causes excluded; anisometropia can produce amblyopia.

Alignment

The cover test reveals a manifest deviation; alternate cover dissociates the eyes to reveal latent deviation.

Red reflex

Leukocoria is an abnormal white reflex and requires urgent ophthalmology assessment.

Link the finding to action

Amblyopia treatment supplies a clear image and may encourage the weaker eye by patching the better eye or using prescribed penalisation. Cataract or pupil-obstructing ptosis can cause deprivation. Childhood leukocoria has lens, vitreous and retinal causes: retinoblastoma is the urgent concern, while persistent fetal vasculature and Coats disease provide important alternative patterns.